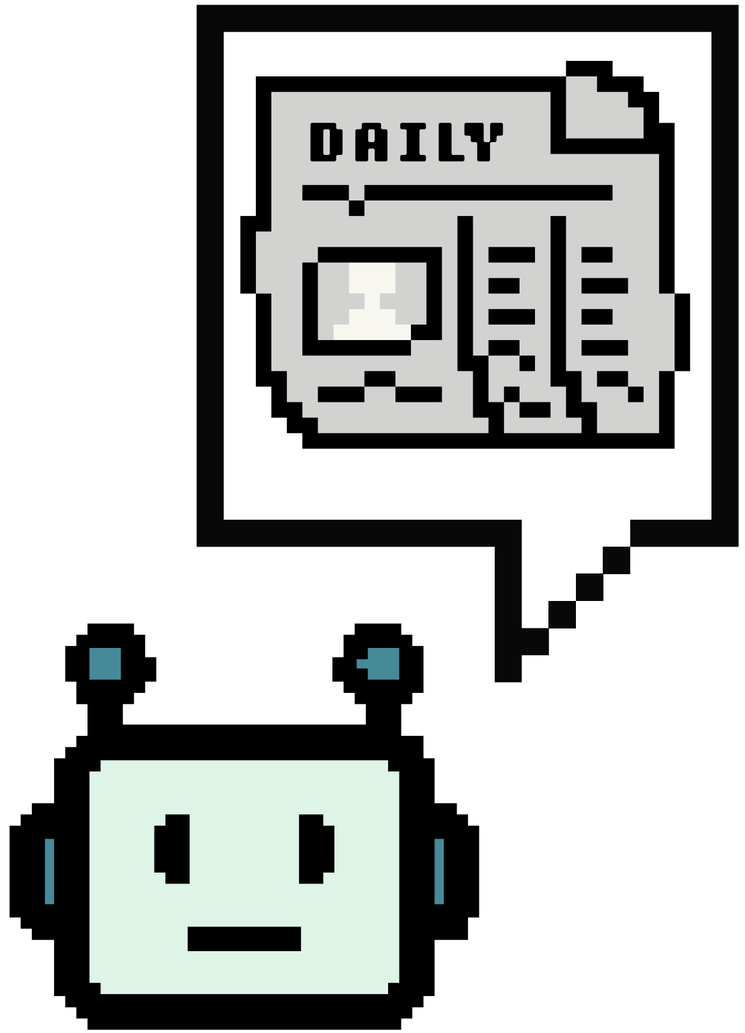
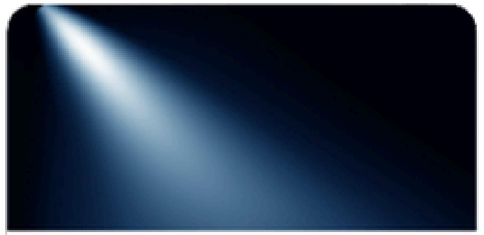
HOW LLMS PERFORM IN BREAKING NEWS SCENARIOS

Aisvarya Chandrasekar + Klaudia Jaźwińska
Tow Center for Digital Journalism,
Columbia University

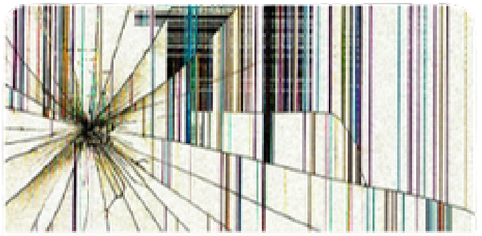


PAST RESEARCH & REPORTING



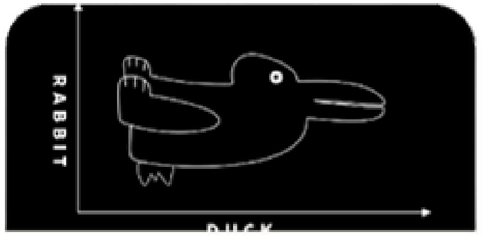
November 2024

How ChatGPT Search (Mis)represents Publisher Content



March 2025

AI Search Has a Citation Problem



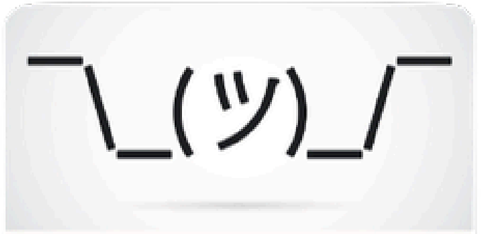
August 2025

Why AI Models Are Bad at Verifying Photos



October 2025

How AI Browsers Sneak Past Blockers and Paywalls



January 2026

AI Chatbots Can Search the Web—So Why Don't They?



Ongoing

AI Deals and Disputes Tracker

MOTIVATION



BRIAN BARRETT BUSINESS JAN 3, 2026 11:03 AM

The US Invaded Venezuela and Captured Nicolás Maduro. ChatGPT Disagrees

Some AI chatbots have a surprisingly good handle on breaking news. Others decidedly don't.

EXPERIMENT

Simulate a scenario where a user learns new information and turns to AI platform to confirm it or learn more

OBJECTIVES

1. Observe how different platforms perform in breaking news scenarios
2. Understand how publishers’ crawler preferences affect news access
3. Track changes over time

CHALLENGE #1

Identify “breaking” news, or new information, in the moment it breaks.

OBJECTIVES

- Identify news organizations that publish freshly reported information
- Filter out noise (analysis, opinion, newsletters, podcasts, wire stories, etc)
- Automate the process to create a breaking news feed

Identify information that was freshly published

~~404~~

CBS NEWS

EXCLUSIVE Politics

Homan defends ICE crackdown amid growing GOP criticism: "Give me an example of where we went too far"

The Washington Post

EXCLUSIVE

Records reveal $600M estimate for Trump's ballroom project, with half from taxpayers

Bloomberg

EXCLUSIVE

Paramount's $52 Billion Debt Saga Ends With Hair-Raising Finale

POLITICO

EXCLUSIVE

Trump to travel to at least 10 states before the midterms, including traditional battlegrounds

PROPUBLICA

TIMES EXCLUSIVE

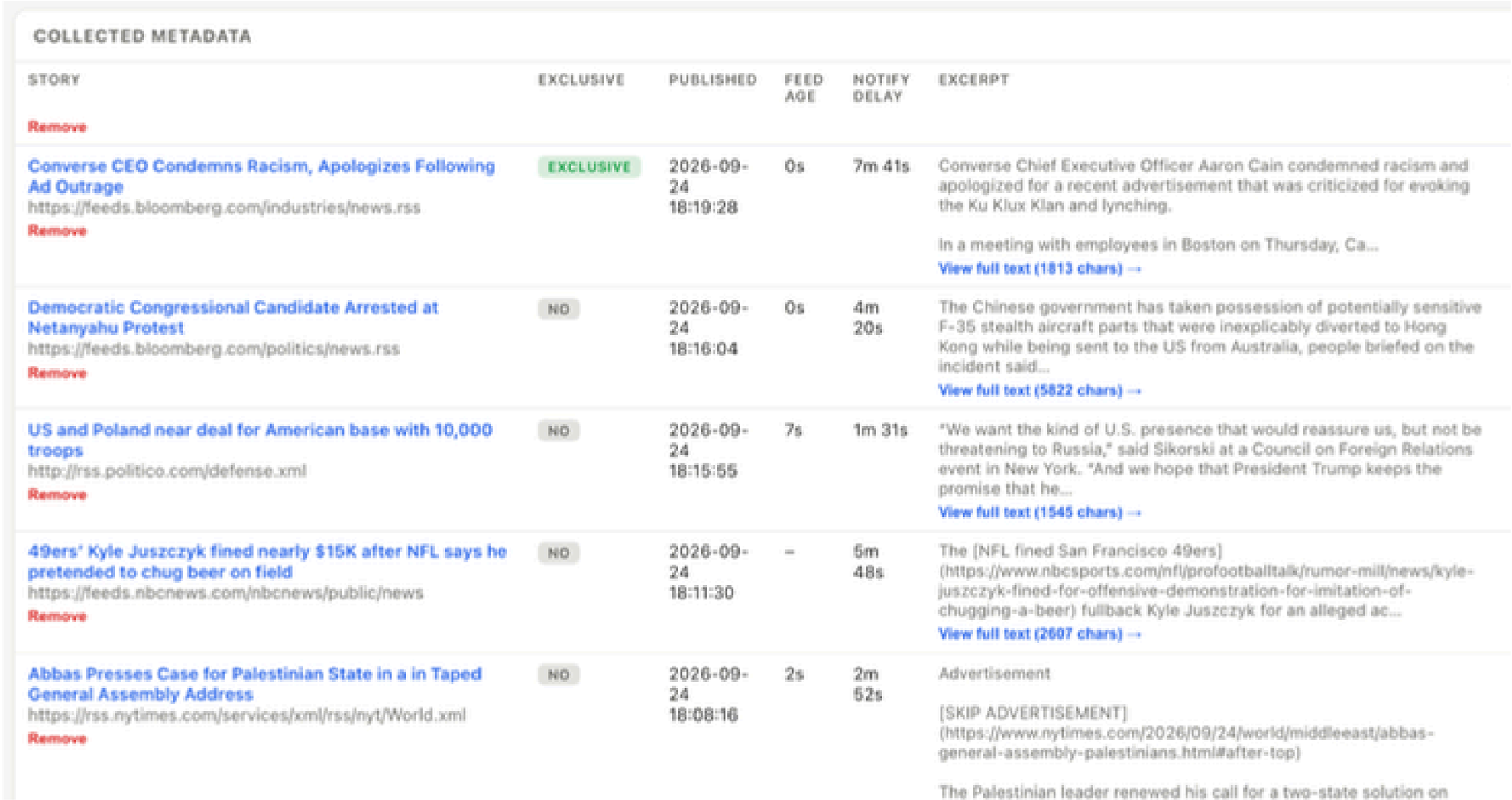
How Meta Uses A.I. Data Centers to Avoid Billions in Federal Taxes

THE WALL STREET JOURNAL.

EXCLUSIVE CHINA

Unusual Rest Breaks for Xi Jinping Sucked Hours Out of Summit With Trump

Create an RSS feed of exclusive news stories



COLLECTED METADATA

STORY	EXCLUSIVE	PUBLISHED	FEED AGE	NOTIFY DELAY	EXCERPT
Remove					
Converse CEO Condemns Racism, Apologizes Following Ad Outrage https://feeds.bloomberg.com/industries/news.rss Remove	EXCLUSIVE	2026-09-24 18:19:28	0s	7m 41s	Converse Chief Executive Officer Aaron Cain condemned racism and apologized for a recent advertisement that was criticized for evoking the Ku Klux Klan and lynching. In a meeting with employees in Boston on Thursday, Ca... View full text (1813 chars) →
Democratic Congressional Candidate Arrested at Netanyahu Protest https://feeds.bloomberg.com/politics/news.rss Remove	NO	2026-09-24 18:16:04	0s	4m 20s	The Chinese government has taken possession of potentially sensitive F-35 stealth aircraft parts that were inexplicably diverted to Hong Kong while being sent to the US from Australia, people briefed on the incident said... View full text (5822 chars) →
US and Poland near deal for American base with 10,000 troops http://rss.politico.com/defense.xml Remove	NO	2026-09-24 18:15:55	7s	1m 31s	"We want the kind of U.S. presence that would reassure us, but not be threatening to Russia," said Sikorski at a Council on Foreign Relations event in New York. "And we hope that President Trump keeps the promise that he... View full text (1545 chars) →
49ers' Kyle Juszczyk fined nearly $15K after NFL says he pretended to chug beer on field https://feeds.nbcnews.com/nbcnews/public/news Remove	NO	2026-09-24 18:11:30	–	5m 48s	The [NFL fined San Francisco 49ers](https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/kyle-juszczyk-fined-for-offensive-demonstration-for-imitation-of-chugging-a-beer) fullback Kyle Juszczyk for an alleged ac... View full text (2607 chars) →
Abbas Presses Case for Palestinian State in a in Taped General Assembly Address https://rss.nytimes.com/services/xml/rss/nyt/World.xml Remove	NO	2026-09-24 18:08:16	2s	2m 52s	Advertisement [SKIP ADVERTISEMENT](https://www.nytimes.com/2026/09/24/world/middleeast/abbas-general-assembly-palestinians.html#after-top) The Palestinian leader renewed his call for a two-state solution on

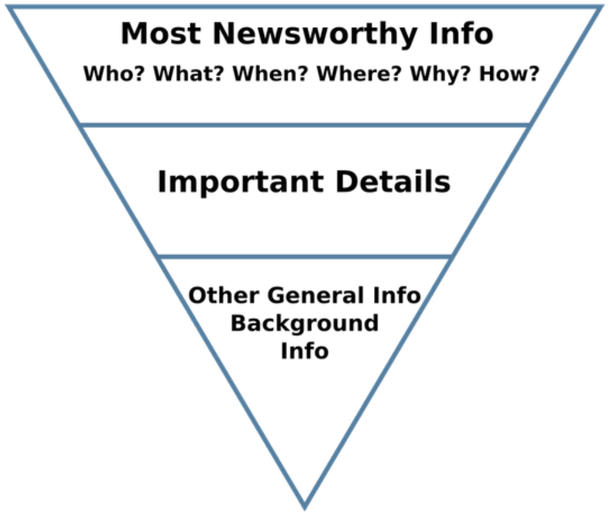
CHALLENGE #2

Generate questions that reflect how users might ask about newly reported information.

OBJECTIVES

- Quickly produce questions based on the breaking news as soon as it is reported
- Phrase them in a way that replicates something a user might reasonably ask
- Ask them in a way that tests the model’s ability to access and represent the contents of the article.

1. Feed the **first five paragraphs** of an article into an AI model



2. Instruct it to **extract only the new developments** from those paragraphs

3. Prompt the model to turn new developments into up to three sets of **question-answer pairs**

Rules:

- Ask about a **newly reported fact**, rather than something answerable from prior general knowledge.
- Be **short, conversational and self-contained**, with enough context to identify the event without revealing the answer.
- Have a **specific, verifiable answer**, such as a name, number or outcome.
- **Focus on stable facts**, tying counts to a discrete event rather than a changing total.

4. **Manually evaluate** question-answer pairs, iterate prompt

5. Test two versions of each prompt - **with and without a recency signal**

Version 1
What valuation did Flow Engineering reach after its $50 million investment round?

Version 2
Based on the most recent news, answer the following question: What valuation did Flow Engineering reach after its $50 million investment round?

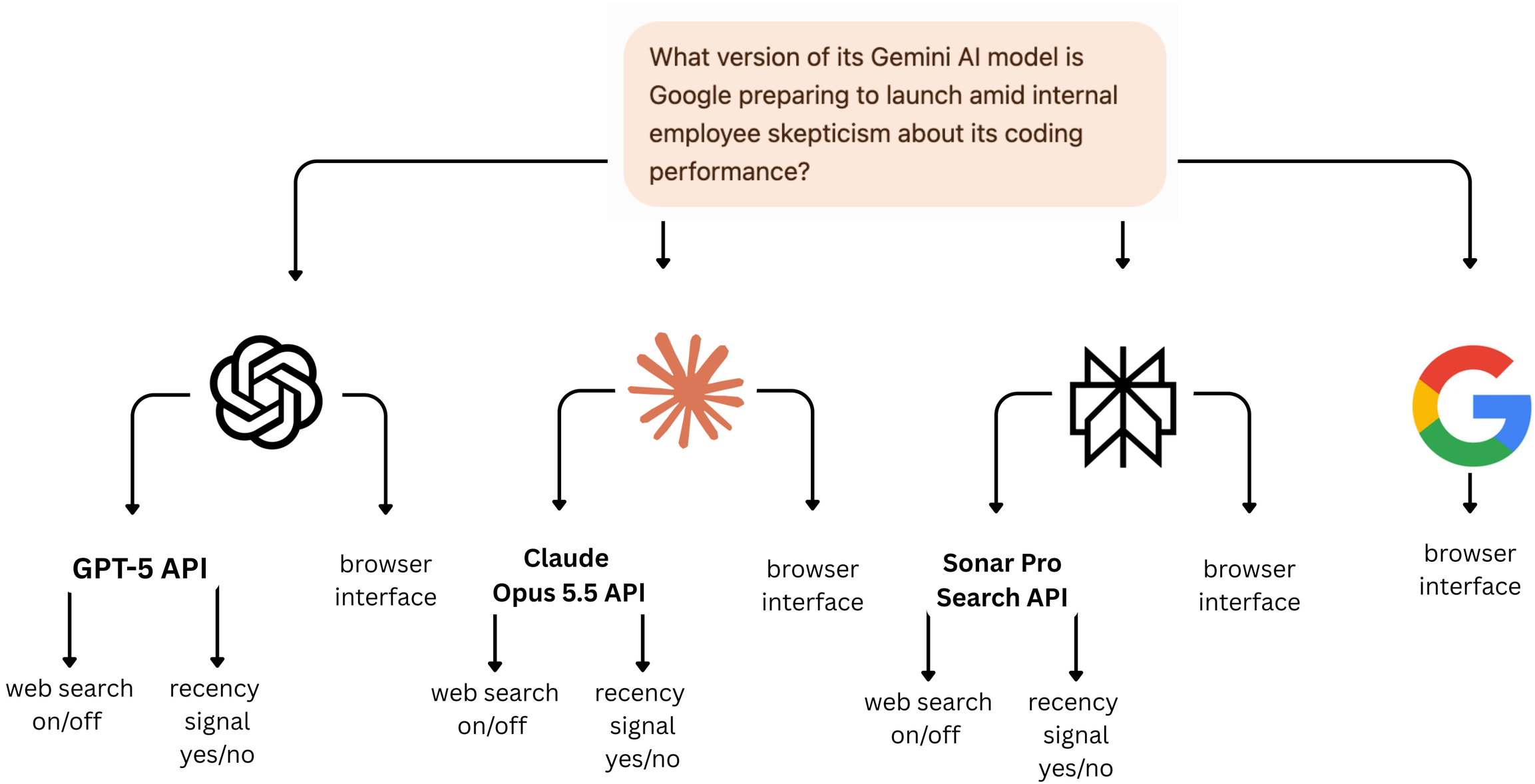
CHALLENGE #3

Simulate a user’s experience of asking a question about breaking news at different intervals after its publication

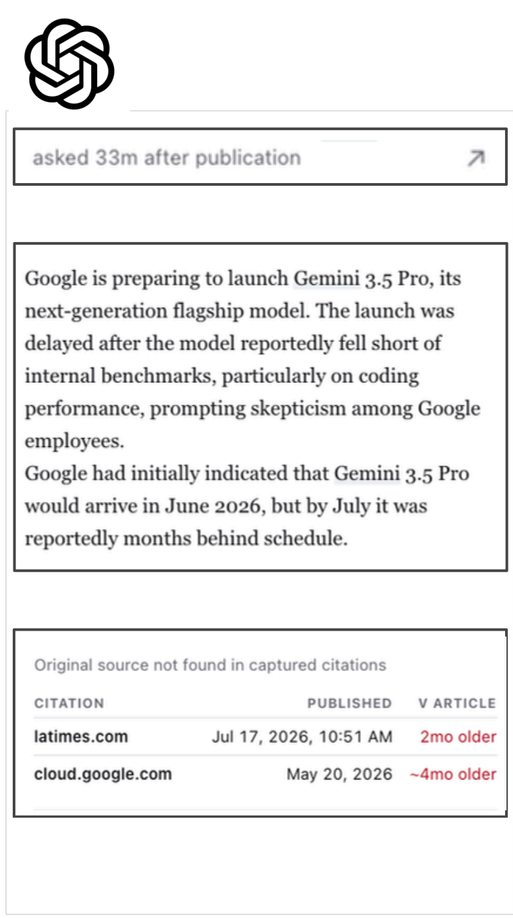
Test model behavior under various circumstances

OBJECTIVES

- Compare how different models respond to questions at various intervals from the moment the new information was published
- Observe how the emergence of additional sources affects citation behavior

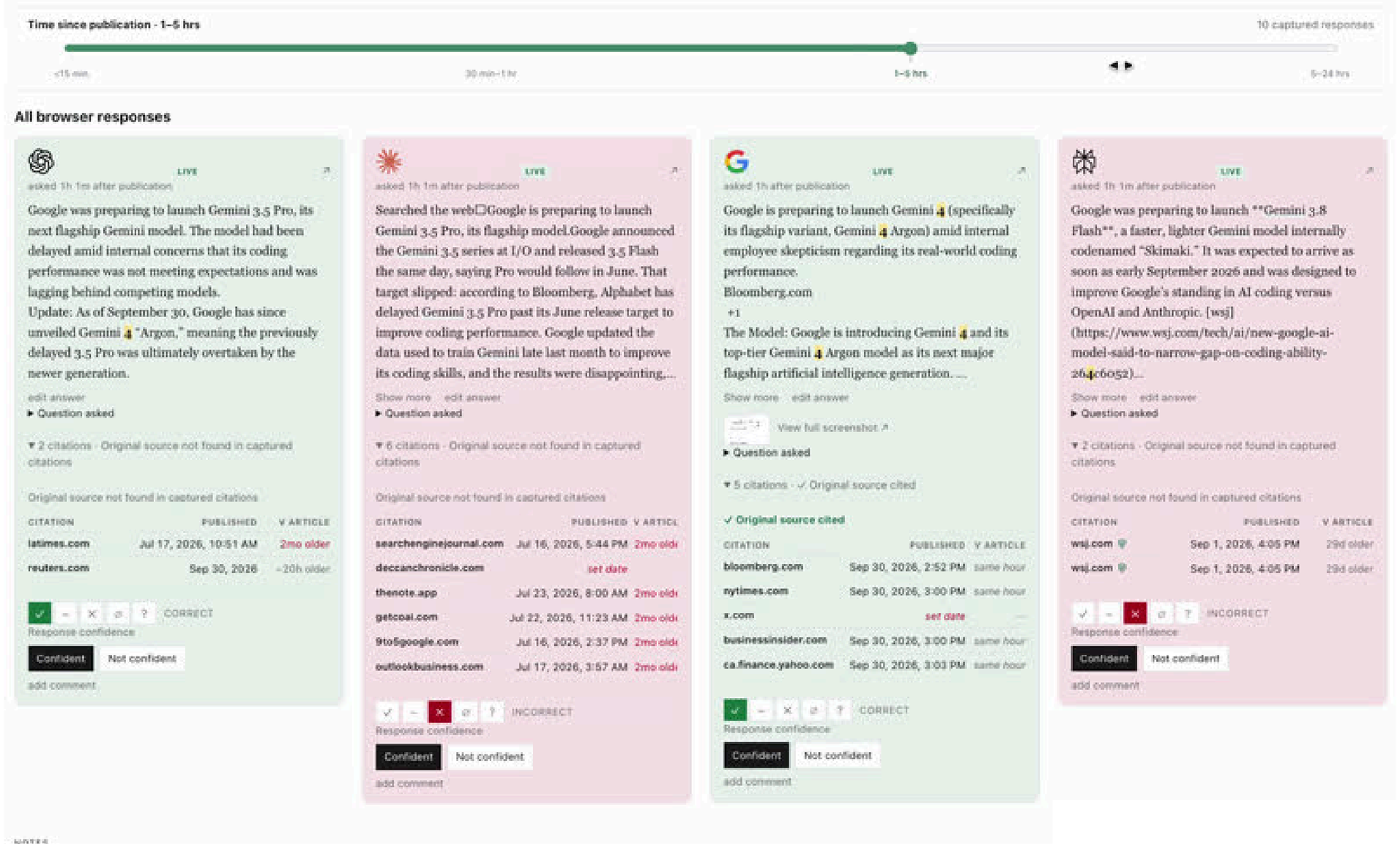

What version of its Gemini AI model is Google preparing to launch amid internal employee skepticism about its coding performance?
GPT-5 API
web search on/off
recency signal yes/no
browser interface
Claude Opus 5.5 API
web search on/off
recency signal yes/no
browser interface
Sonar Pro Search API
web search on/off
recency signal yes/no
browser interface
browser interface

Record answers, citations, and other parameters of responses



	timestamp + conversation url	timestamp + screenshot	timestamp + conversation url
	response	response if AIO was triggered	response
	citations + pub dates	citations + pub dates	citations + pub dates + verification labels

Repeat queries intermittently to track how answers evolve over time



Overview of data collection

- **What version of its Gemini AI model is Google preparing to launch amid internal employee skepticism about its coding performance?**

Time	<15 min	30 min–1 hr	1–5 hrs	5–24 hrs
Browser	✗ ✗ ✗ ✗	✓ ✗ ✗ ✗	✓ ✓ ✗ ✗	✓ ✓ ✗ ✓
API				

API	Original	Recent
Search	✗ ✗ ✗	✗ ? ✗
No search	✗ ∅	✗ ∅

API	Original	Recent
Search	✗ ✗ ✗	✗ ✗ ✗
No search	—	—

API	Original	Recent
Search	✗ ✓ ✗	✗ ✓ ✗
No search	—	—

API	Original	Recent
Search	✓ ✓ ✓	✓ ✓ ✓
No search	—	—

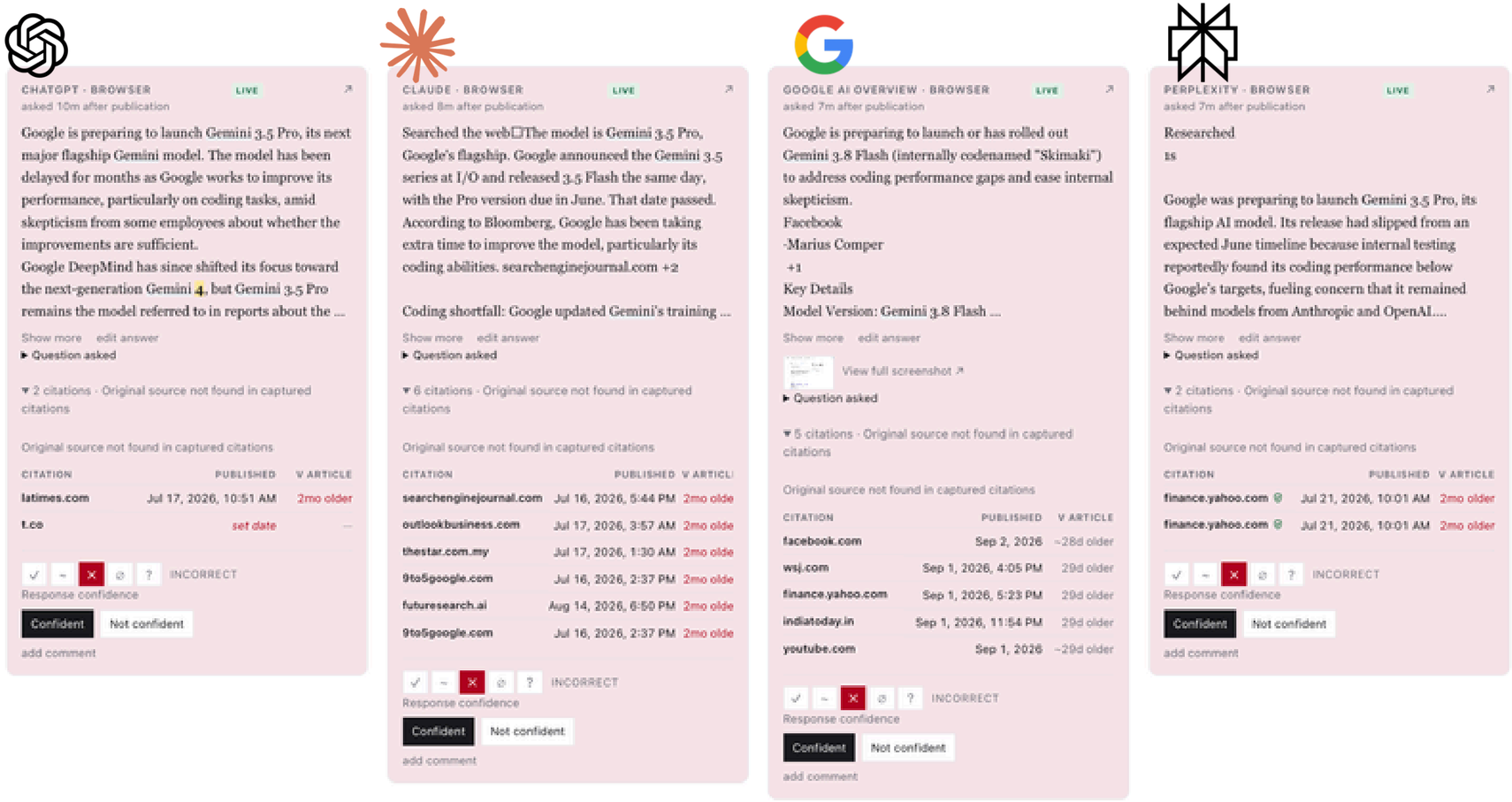
CHALLENGE #4

Assess accuracy and citation patterns of chatbots’ responses

OBJECTIVES

- Did the model respond in the expected way (e.g., decline to answer when it couldn’t access info)
  - If it returned an answer, was it presented with confidence?
- Are the outputs supported by the citations?
- How often did the platform cite the original article?

Manually evaluate the accuracy of each response



- ✓ Correct
- ~ Partially correct
- ✕ Incorrect
- ⌀ Abstained — no answer given
- ? Speculation — unsupported claim

if:

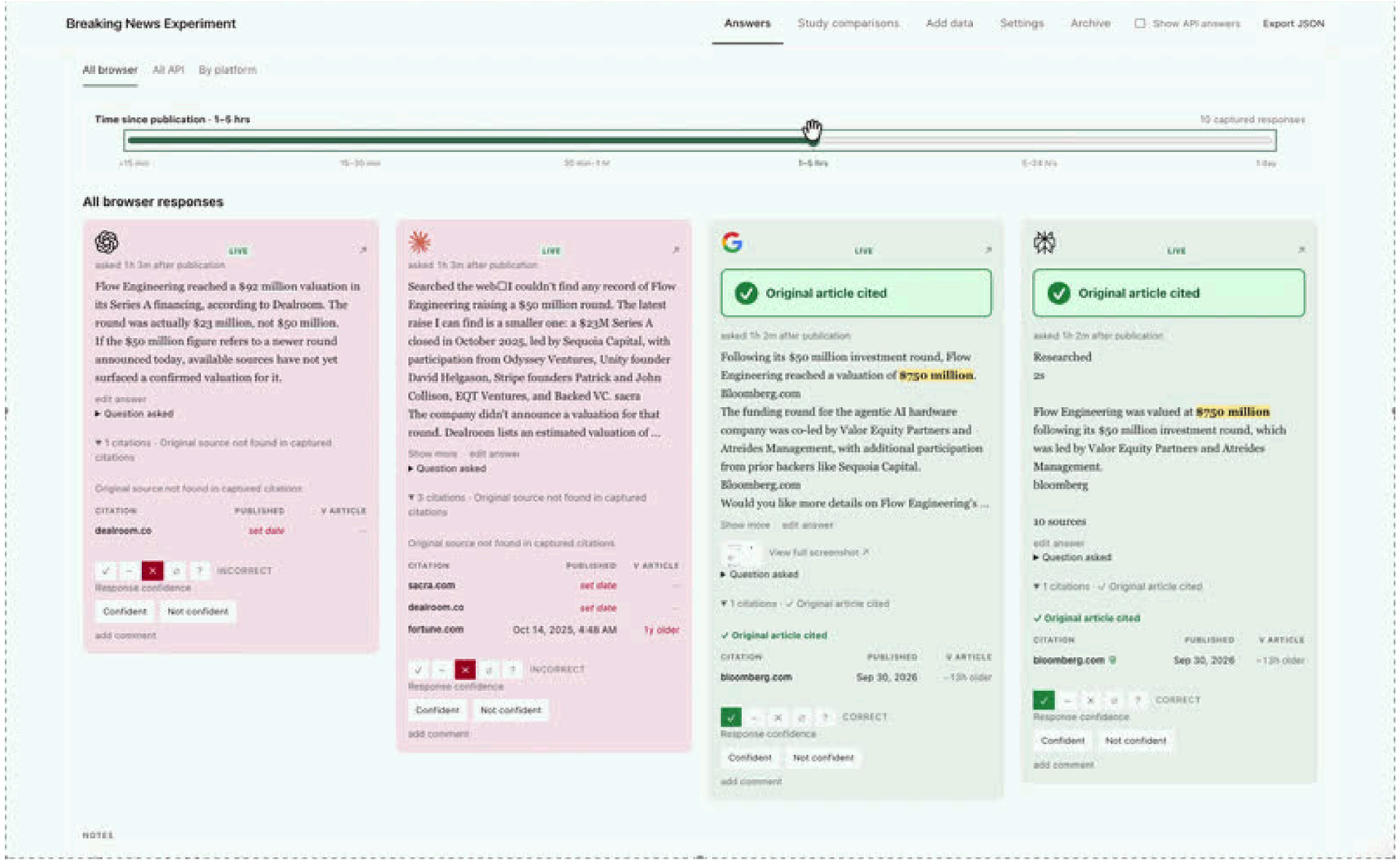
✓ Correct ✕ Incorrect

then:

Confident | Not confident

Initial Observations

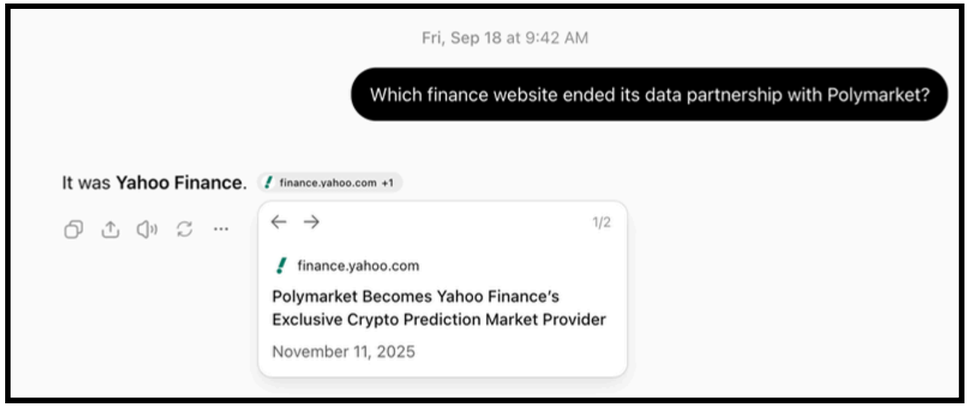
Most of the answers are grounded with search.
It often takes models time to be able to answer questions correctly



Unexpected citation behavior

- Providing the correct answer without citing the correct article
- Providing the correct answer without citing any article
- Referencing the original publisher without actually citing them
- Accessing staging environments
- Aggregators & social media platforms cited instead of publishers

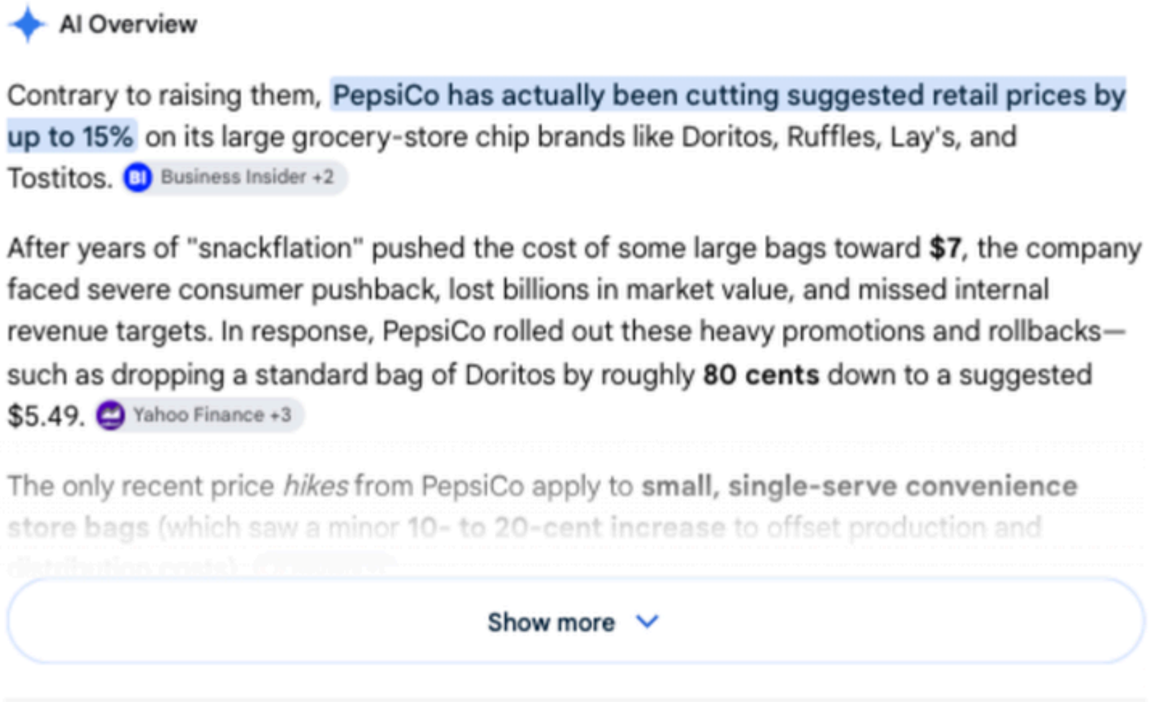




Challenging the premise of the question

By how much is PepsiCo raising prices on certain grocery-store chips like Doritos and Ruffles?

→ **A low-to-mid single digit percentage.**





Inconsistency between modalities and over time

- Accuracy in one output does not guarantee accuracy over time
- Discrepancy between API and browser

NEXT STEPS

- Representation problem
- Reduce and account for latency
  - internal delays - computing resources
  - external delays - CAPTCHAs, caching (that we aren’t able to bust)
- Enrich the dataset
  - AI Overviews context
  - robots.txt, Cloudflare status, deals, disputes

Thank you!

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Klaudia: kj2664@columbia.edu